

File permissions in Linux

Project description

In this project, I acted as a security professional to ensure users on a research team had the correct authorization and permissions on the network. I used Linux commands to examine the current file system permissions, identify vulnerabilities (such as unauthorized write access), and used the `chmod` command to modify file and directory permissions. The goal was to align system authorization with the organization's security policies by applying the principle of least privilege.

Check file and directory details

To check the current permissions of all files and directories, including hidden files, I used the following command: `ls -la`

```
researcher2@007330c5d47c:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 May  4 14:58 .
drwxr-xr-x 3 researcher2 research_team 4096 May  4 15:04 ..
-rw--w---- 1 researcher2 research_team  46 May  4 14:58 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 May  4 14:58 drafts
-rw-rw-rw- 1 researcher2 research_team  46 May  4 14:58 project_k.txt
-rw-r----- 1 researcher2 research_team  46 May  4 14:58 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May  4 14:58 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May  4 14:58 project_t.txt
```

Describe the permissions string

Taking the file `project_k.txt` as an example, the permissions string is `-rw-rw-rw-`. This 10-character string is divided into four sections:

1. **The first character (-):** Indicates the file type. A hyphen - means it is a regular file (if it were a d, it would be a directory).
2. **The next three characters (rw-):** Represent the permissions for the **User** (owner of the file). In this case, the user has read (r) and write (w) permissions, but no execute (-) permission.
3. **The following three characters (rw-):** Represent the permissions for the **Group**. The group also has read and write permissions.
4. **The last three characters (rw-):** Represent the permissions for **Other** (anyone else on the system). They can also read and write to this file.

Change file permissions

The organization does not allow "others" to have write access to any files. Upon analyzing the list, I identified that the project_k.txt file had read and write permissions for "others" (Other = read, write). To fix this and remove the write permission, I used the following command: `chmod o-w project_k.txt`

```
researcher2@007330c5d47c:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 May  4 14:58 .
drwxr-xr-x 3 researcher2 research_team 4096 May  4 15:04 ..
-rw--w---- 1 researcher2 research_team  46 May  4 14:58 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 May  4 14:58 drafts
-rw-rw-r-- 1 researcher2 research_team  46 May  4 14:58 project_k.txt
-rw-r----- 1 researcher2 research_team  46 May  4 14:58 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May  4 14:58 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May  4 14:58 project_t.txt
```

Change file permissions on a hidden file

The hidden file .project_x.txt should not have write permissions for anyone, but the user and group should be able to read it. Originally, the user had read and write permissions, and the group only had write permission (-rw--w----). To set the correct permissions (read-only for user and group, and none for others), I used the command: `chmod u-w,g+r-w .project_x.txt`

```
researcher2@007330c5d47c:~/projects$ chmod u-w,g+r-w .project_x.txt
researcher2@007330c5d47c:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 May  4 14:58 .
drwxr-xr-x 3 researcher2 research_team 4096 May  4 15:04 ..
-r--r----- 1 researcher2 research_team  46 May  4 14:58 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 May  4 14:58 drafts
-rw-rw-r-- 1 researcher2 research_team  46 May  4 14:58 project_k.txt
-rw-r----- 1 researcher2 research_team  46 May  4 14:58 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May  4 14:58 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May  4 14:58 project_t.txt
```

Change directory permissions

The drafts directory belongs to the researcher2 user, and only they should be allowed to access it. The initial permissions for the directory were drwx--x---, which means the group had execute permission (access to enter the directory). To remove this access from the group, I used the command: `chmod g-x drafts`

```
researcher2@007330c5d47c:~/projects$ chmod g-x drafts
researcher2@007330c5d47c:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 May  4 14:58 .
drwxr-xr-x 3 researcher2 research_team 4096 May  4 15:04 ..
-r--r----- 1 researcher2 research_team  46 May  4 14:58 .project_x.txt
drwx----- 2 researcher2 research_team 4096 May  4 14:58 drafts
-rw-rw-r-- 1 researcher2 research_team  46 May  4 14:58 project_k.txt
-rw-r----- 1 researcher2 research_team  46 May  4 14:58 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May  4 14:58 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May  4 14:58 project_t.txt
```

Summary

To protect the research team's data, I started the analysis with the `ls -la` command to map all file permissions, including hidden system files. I identified critical vulnerabilities, such as the `project_k.txt` file being open for modification by any external user, and used the `chmod o-w` command to mitigate that risk. Finally, I revoked execute access on confidential directories and turned hidden archive files (`.project_x.txt`) into read-only files, ensuring that the current file system strictly complies with the organization's access control security policies.